

Perfectoid Moduli of Quotient Bundles on the Fargues–Fontaine Curve

Abstract

Let $F/\mathbf{Q}_p$ be a finite extension and let C be a complete algebraically closed perfectoid field in the usual Fargues–Fontaine setting. For integers $1 \leq d < n$, let $W_{n,d}$ be the moduli v -sheaf of quotient bundles

$$\mathcal{O}_{X_S}(1/n) \twoheadrightarrow Q$$

on relative Fargues–Fontaine curves whose geometric fibres have bundle type $\mathcal{O}(1/d)$. We prove that $W_{n,d}$ is represented by a perfectoid space.

1 Setup and statement

Let $F/\mathbf{Q}_p$ be a finite extension, and let C be a complete algebraically closed perfectoid extension of F . For a perfectoid space S over C , write X_S for the relative Fargues–Fontaine curve attached to $S^\flat$ and F , and put

$$X := X_{\mathrm{Spa}(C, C^+)}.$$

For $\lambda \in \mathbf{Q}$, write $\mathcal{O}_{X_S}(\lambda)$ for the standard vector bundle of slope λ .

Convention 1.1 (Global and geometric bundle type). For vector bundles E, E' on X_S , the notation $E \cong_{X_S} E'$ means a global isomorphism on the whole relative curve. The notation

$$E \cong_{\mathrm{geom}/S} \mathcal{O}(\lambda)^{\oplus m}$$

means that for every geometric point $\bar{s} \rightarrow S$ one has a global isomorphism

$$E|_{X_{\bar{s}}} \cong_{X_{\bar{s}}} \mathcal{O}_{X_{\bar{s}}}(\lambda)^{\oplus m}.$$

For bundles on the absolute curve we write $\cong_X$. The symbol $\simeq$ is reserved for isomorphisms of functors, v -sheaves, diamonds, spaces, or categories.

Fix integers $1 \leq d < n$ and set

$$V := \mathcal{O}_X(1/n), \quad r := n - d.$$

Define the set-valued v -sheaf $W_{n,d}$ by

$$W_{n,d}(S) = \left\{ \begin{array}{l} \text{surjections of vector bundles } q : V_S \twoheadrightarrow Q \\ \text{such that } Q \cong_{\mathrm{geom}/S} \mathcal{O}(1/d) \end{array} \right\} / \cong.$$

An isomorphism of two quotient maps is required to commute with the maps from V_S ; since the quotient maps are surjective, there are no nontrivial automorphisms.

Theorem 1.2. *For every $1 \leq d < n$, the v -sheaf $W_{n,d}$ is represented by a perfectoid space over C .*

2 Formal reductions

2.1 Open-local representability

Lemma 2.1 (Open-local representability). *Let Y be a diamond. Suppose that every point $y \in |Y|$ is contained in an open subdiamond $U_y \subset Y$ represented by a perfectoid space. Then Y is represented by a perfectoid space.*

Proof. By Scholze’s correspondence between open subdiamonds and open subsets of the underlying topological space [4, Proposition 11.15], the U_y form an open covering of Y . Since this cover is represented by perfectoid spaces, Y itself is represented by a perfectoid space. $\square$

2.2 Kernels and the diamond structure

Lemma 2.2 (Kernel formulation). *The v -sheaf $W_{n,d}$ is naturally isomorphic to the v -sheaf assigning to S/C the stably injective maps*

$$K \longrightarrow V_S, \quad K \cong_{\text{geom}/S} \mathcal{O}^{\oplus r}.$$

Here stable injectivity means injectivity after every base change, equivalently after pullback to every geometric point, as in the discussion preceding [3, Lemma A.2.2].

Proof. Let $q : V_S \rightarrow Q$ be an S -point of $W_{n,d}$ and put $K = \ker(q)$. On a sufficiently small Tate affinoid open $U = \text{Spa}(R, R^+) \subset X_S$, vector bundles correspond to finite projective R -modules by Kedlaya–Liu [5, Theorem 2.7.7]. Shrinking U so that local basis sections of Q lift through q , the restriction of q is a surjection $M \twoheadrightarrow N$ of finite projective modules. Since N is projective, this surjection splits; its kernel is finite projective and the resulting short exact sequence remains exact after every base change. Thus K is a vector bundle and $K \hookrightarrow V_S$ is stably injective. We will use the same local splitting argument below whenever an injection has vector-bundle cokernel. At a geometric point $\bar{s} \rightarrow S$, the bundle $K_{\bar{s}}$ has rank r and degree 0.

The Harder–Narasimhan filtration on the absolute Fargues–Fontaine curve splits, and the stable bundle of a reduced positive slope a/b has rank b [2, Theorem I.3.2]. If $K_{\bar{s}}$ had a positive stable summand $\mathcal{O}(a/b)$, then $b \leq r < n$, hence

$$\frac{a}{b} \geq \frac{1}{b} > \frac{1}{n}.$$

The restriction of $K_{\bar{s}} \hookrightarrow \mathcal{O}(1/n)$ to that summand would be a nonzero section of the semistable bundle $\underline{\text{Hom}}(\mathcal{O}(a/b), \mathcal{O}(1/n))$, whose slope is $1/n - a/b < 0$, contradicting [2, Proposition II.2.5(i)]. Thus all slopes of $K_{\bar{s}}$ are nonpositive. Since its total degree is zero, all slopes are zero, and therefore

$$K_{\bar{s}} \cong \mathcal{O}^{\oplus r}.$$

Conversely, if $K \rightarrow V_S$ is stably injective and $K \cong_{\text{geom}/S} \mathcal{O}^{\oplus r}$, then [3, Lemma A.2.2] shows that V_S/K is a vector bundle and

$$V_S/K \cong_{\text{geom}/S} \mathcal{O}\left(\frac{1}{n-r}\right) = \mathcal{O}(1/d).$$

The two constructions are inverse and commute with base change. $\square$

Proposition 2.3. *The v -sheaf $W_{n,d}$ is a diamond.*

Proof. Consider the sheaf of framed injections

$$W_{n,d}^f := \text{Inj}(\mathcal{O}_X^{\oplus r}, V).$$

By [1, Proposition 3.3.6], this is an open, locally spatial subdiamond of $\text{Hom}(\mathcal{O}_X^{\oplus r}, V)$. By [3, Lemma A.2.2], each framed injection gives an object of $W_{n,d}$.

Let $G = \text{GL}_r(F)$ be the locally profinite group v -sheaf. It acts freely by precomposition. By [2, Corollary II.2.20], every geometric-slope-zero kernel is the vector bundle attached to a pro-étale F -local system; its sheaf of frames is therefore a G -torsor and is pro-étale locally nonempty. Consequently

$$W_{n,d} \simeq W_{n,d}^f / G$$

as v -sheaves. More precisely, for every perfectoid $S \rightarrow W_{n,d}$, the pullback $W_{n,d}^f \times_{W_{n,d}} S$ is the sheaf of frames of the F -local system corresponding to the kernel. It is a G -torsor in the sense of [4, Definition 10.12]. By [4, Lemma 10.13], this pullback is represented by a perfectoid space and is pro-étale and surjective over S . Hence $W_{n,d}^f \rightarrow W_{n,d}$ is a surjective pro-étale morphism. Since $W_{n,d}^f$ is a diamond, [4, Proposition 11.6] implies that $W_{n,d}$ is a diamond. $\square$

2.3 The transverse open locus

Fix a stably injective map on the absolute curve

$$t : \mathcal{O}_X \hookrightarrow V.$$

For an S -point $K \hookrightarrow V_S$ of $W_{n,d}$, let

$$(K, t)_S : K \oplus \mathcal{O}_{t,S} \longrightarrow V_S$$

denote the sum of the two inclusions. Define

$$U_t(S) := \{K \hookrightarrow V_S \in W_{n,d}(S) \mid (K, t)_S \text{ is stably injective}\}.$$

Lemma 2.4 (Openness of the transverse locus). *The inclusion $U_t \hookrightarrow W_{n,d}$ is an open immersion of diamonds.*

Proof. Use the framed G -torsor $W_{n,d}^f \rightarrow W_{n,d}$ from Proposition 2.3. On a perfectoid test space S/C , the pullback of U_t is

$$\tilde{U}_t(S) = \left\{ f : \mathcal{O}_{X_S}^{\oplus r} \rightarrow V_S \mid \begin{array}{l} (f, t_S) : \mathcal{O}_{X_S}^{\oplus(r+1)} \rightarrow V_S \\ \text{is fibrewise injective} \end{array} \right\}.$$

Consider the morphism

$$\Phi_t : \text{Hom}(\mathcal{O}_X^{\oplus r}, V) \longrightarrow \text{Hom}(\mathcal{O}_X^{\oplus(r+1)}, V), \quad f \longmapsto (f, t).$$

Then

$$\tilde{U}_t = W_{n,d}^f \cap \Phi_t^{-1}(\text{Inj}(\mathcal{O}_X^{\oplus(r+1)}, V)).$$

The injective locus is open by [1, Proposition 3.3.6], so $\tilde{U}_t$ is open in $W_{n,d}^f$. Since openness is v -local on the target by [4, Proposition 10.11(i)], $U_t \hookrightarrow W_{n,d}$ is open. $\square$

3 The two-step reduction attached to a section

Keep a fixed stably injective $t : \mathcal{O}_X \hookrightarrow V$. By [3, Lemma A.2.2],

$$V^t := V/\mathcal{O}_t$$

is a vector bundle of rank $n-1$, degree 1, and geometric type $\mathcal{O}(1/(n-1))$. Since it is a bundle on the absolute curve, the classification in [2, Theorem I.3.2] gives a global isomorphism

$$\alpha_t : V^t \xrightarrow{\sim}_X \mathcal{O}_X(1/(n-1)), \quad (1)$$

which we choose once and for all.

3.1 The incidence diamond inside $W_{n,d-1}$

Assume $d > 1$. Define a subfunctor

$$\mathcal{V}_t \subset W_{n,d-1}$$

by declaring that $\widetilde{K} \hookrightarrow V_S$ lies in $\mathcal{V}_t(S)$ if t_S factors through $\widetilde{K}$ and the quotient $\widetilde{K}/\mathcal{O}_{t,S}$ is a vector bundle. In particular, the induced inclusion $\mathcal{O}_{t,S} \hookrightarrow \widetilde{K}$ is locally split and remains injective after every base change, by the argument in Lemma 2.2.

Proposition 3.1 (The incidence reduction). *There is a natural equivalence*

$$\mathcal{V}_t \simeq W_{n-1,d-1}.$$

With the choice (1), it sends

$$\widetilde{K} \hookrightarrow V_S \mapsto J := \widetilde{K}/\mathcal{O}_{t,S} \hookrightarrow V_S^t \xrightarrow{\alpha_t} \mathcal{O}_{X_S}(1/(n-1)).$$

Proof. If $\widetilde{K} \in \mathcal{V}_t(S)$, then $\widetilde{K} \cong_{\text{geom}/S} \mathcal{O}^{\oplus(r+1)}$ because it is the kernel of an object of $W_{n,d-1}$. At a geometric point $\bar{s} \rightarrow S$, full faithfulness in [2, Corollary II.2.20] identifies the inclusion

$$\mathcal{O}_{X_{\bar{s}}} \hookrightarrow \widetilde{K}_{\bar{s}} \cong \mathcal{O}_{X_{\bar{s}}}^{\oplus(r+1)}$$

with an injective F -linear map $F \hookrightarrow F^{r+1}$. After an element of $\text{GL}_{r+1}(F)$ it is the first coordinate inclusion, and therefore

$$J = \widetilde{K}/\mathcal{O}_{t,S} \cong_{\text{geom}/S} \mathcal{O}^{\oplus r}.$$

Moreover,

$$V_S^t/J \cong V_S/\widetilde{K} \cong_{\text{geom}/S} \mathcal{O}(1/(d-1)).$$

Thus $J \hookrightarrow V_S^t$ is an object of the kernel formulation of $W_{n-1,d-1}$ from Lemma 2.2.

Conversely, let $J \hookrightarrow V_S^t$ represent an S -point of $W_{n-1,d-1}$ and define the inverse image

$$\widetilde{K} := V_S \times_{V_S^t} J.$$

This is the kernel of the surjection $V_S \rightarrow V_S^t/J$ between vector bundles, so it is a vector bundle by the local splitting argument in Lemma 2.2. The sequence

$$0 \longrightarrow \mathcal{O}_{t,S} \longrightarrow \widetilde{K} \longrightarrow J \longrightarrow 0 \quad (2)$$

is locally split and remains exact after every base change, since J is a vector bundle. Moreover,

$$V_S/\widetilde{K} \cong V_S^t/J \cong_{\text{geom}/S} \mathcal{O}(1/(d-1)).$$

At a geometric point $\bar{s} \rightarrow S$, (2) splits because

$$\mathrm{Ext}_{X_{\bar{s}}}^1(\mathcal{O}^{\oplus r}, \mathcal{O}) = H^1(X_{\bar{s}}, \mathcal{O})^{\oplus r} = 0$$

by [2, Proposition II.2.5(ii)]. Hence

$$\widetilde{K} \cong_{\mathrm{geom}/S} \mathcal{O}^{\oplus(r+1)}.$$

Since its quotient in V_S is a vector bundle, the same local splitting argument shows that $\widetilde{K} \hookrightarrow V_S$ is stably injective and defines a point of $\mathcal{V}_t$. The two constructions are inverse and compatible with base change. $\square$

3.2 The reduction morphism

For $K \hookrightarrow V_S$ in $U_t(S)$, the map

$$K \oplus \mathcal{O}_{t,S} \longrightarrow V_S$$

is stably injective. Its image, denoted

$$\widetilde{K}_t(K) := K + \mathcal{O}_{t,S},$$

is globally isomorphic to $K \oplus \mathcal{O}_{t,S}$. By [3, Lemma A.2.2],

$$V_S/\widetilde{K}_t(K) \cong_{\mathrm{geom}/S} \mathcal{O}\left(\frac{1}{n - (r+1)}\right) = \mathcal{O}(1/(d-1)).$$

Define

$$\sigma_{(t)} : U_t \longrightarrow W_{n,d-1}, \quad K \longmapsto K + \mathcal{O}_t. \quad (3)$$

Its image lies in the incidence subdiamond $\mathcal{V}_t$. Writing ρ_t for the resulting factorization gives

$$\begin{array}{ccc} U_t & \xrightarrow{\sigma_{(t)}} & W_{n,d-1} \\ & \searrow \rho_t & \uparrow \\ & \mathcal{V}_t & \xrightarrow{\sim} W_{n-1,d-1}. \end{array} \quad (4)$$

3.3 The splitting torsor

Throughout this subsection, assume that $W_{n-1,d-1}$ is represented by a perfectoid space. By Proposition 3.1, the base $\mathcal{V}_t$ is then perfectoid as well.

Let $J_t^{\mathrm{univ}} \hookrightarrow V_{\mathcal{V}_t}^t$ denote the universal rank- r kernel on $X_{\mathcal{V}_t}$, and let $\widetilde{K}_t^{\mathrm{univ}}$ be its inverse image in $V_{\mathcal{V}_t}$. There is a universal short exact sequence

$$e_t : 0 \longrightarrow \mathcal{O}_{t,\mathcal{V}_t} \longrightarrow \widetilde{K}_t^{\mathrm{univ}} \longrightarrow J_t^{\mathrm{univ}} \longrightarrow 0. \quad (5)$$

Write $\mathrm{Spl}_{\mathcal{V}_t}(e_t)$ for its sheaf of splittings.

Proposition 3.2 (The chart is a splitting torsor). *The factored morphism $\rho_t : U_t \rightarrow \mathcal{V}_t$ induces a natural equivalence over $\mathcal{V}_t$*

$$U_t \simeq \mathrm{Spl}_{\mathcal{V}_t}(e_t).$$

The splitting sheaf is a torsor under the relative Hom sheaf

$$\mathcal{H}_t(S) = \mathrm{Hom}_{X_S}(J_{t,S}^{\mathrm{univ}}, \mathcal{O}_{t,S}), \quad S \longrightarrow \mathcal{V}_t,$$

and $\mathcal{H}_t$ is a pro-étale F -local system of rank r .

Proof. For $K \in U_t(S)$, the equality

$$\widetilde{K}_t(K) = K \oplus_{X_S} \mathcal{O}_{t,S}$$

gives a splitting of the pullback of (5). Conversely, a splitting

$$s : J \longrightarrow \widetilde{K}$$

of the extension attached to an S -point $\widetilde{K} \in \mathcal{V}_t(S)$ has image $K = \text{im}(s)$ and gives

$$\widetilde{K} = \mathcal{O}_{t,S} \oplus_{X_S} K.$$

In particular, $K \cong_{X_S} J$ is a vector bundle and $K \cong_{\text{geom}/S} \mathcal{O}^{\oplus r}$. The inclusion $K \hookrightarrow \widetilde{K}$ is split, and $\widetilde{K} \hookrightarrow V_S$ has vector-bundle cokernel, so the local splitting argument in Lemma 2.2 shows that both inclusions remain injective after every base change. Thus $K \hookrightarrow V_S$ and $K \oplus \mathcal{O}_{t,S} \rightarrow V_S$ are stably injective. Applying [3, Lemma A.2.2] to $K \hookrightarrow V_S$ gives

$$V_S/K \cong_{\text{geom}/S} \mathcal{O}(1/d),$$

so K belongs to $U_t(S)$. These constructions are inverse.

For any short exact sequence, the difference of two splittings is a unique map from the quotient to the kernel. Hence $\text{Spl}_{\mathcal{V}_t}(e_t)$ is a torsor under $\mathcal{H}_t$ once splittings exist locally. All three terms in (5) have constant Harder–Narasimhan polygon zero. The equivalence of [2, Corollary II.2.20] identifies this exact sequence of vector bundles with an exact sequence of pro-étale F -local systems: exactness can be checked on geometric fibres, and these remain exact by local splitting of (5). After a pro-étale cover the three local systems are constant. Their maps are continuous matrices with entries in F ; near any point, a nonzero maximal minor of the quotient map stays invertible and gives a continuous right inverse. Thus the sequence, and hence (5), splits pro-étale locally. Under the same equivalence, $\mathcal{H}_t$ is the dual of the rank- r local system corresponding to J_t^{univ} , hence is a rank- r F -local system. $\square$

Lemma 3.3 (Torsors under an F -local system). *Let B be a perfectoid space, let L be a finite-rank pro-étale F -local system on B , and let $P \rightarrow B$ be a pro-étale torsor under L . Then P is represented by a perfectoid space, pro-étale over B .*

Proof. The assertion is local for open covers of B . Regard L as a $\mathbf{Q}_p$ -local system. By [5, Proposition 8.4.6 and Lemma 9.1.11], after passing to an affinoid open cover of B we may choose a $\mathbf{Z}_p$ -local system $\Lambda \subset L$ with $L = \Lambda[1/p]$. We work on one such open.

For $k \geq 0$, form the quotient v -sheaf

$$P_k := P/(p^k \Lambda).$$

Choose a common pro-étale cover trivializing Λ and the torsor P . On that cover, P_0 is a disjoint union of copies of the base, indexed by $\mathbf{Q}_p^N/\mathbf{Z}_p^N$, where $N = \text{rank}_{\mathbf{Z}_p} \Lambda$. The separated étale descent theorem [4, Proposition 9.7], which applies over an arbitrary perfectoid base, therefore represents P_0 by a perfectoid space separated and étale over B . Similarly, on the same cover, $P_{k+1} \rightarrow P_k$ has fibre $p^k \mathbf{Z}_p^N/p^{k+1} \mathbf{Z}_p^N$. Finite étale descent in [4, Proposition 9.7] shows that all these transition maps are finite étale.

There is a natural isomorphism of v -sheaves

$$P \simeq \varprojlim_{k \geq 0} P_k.$$

Indeed, this may be checked after the chosen cover, where it is the isomorphism of topological groups $\mathbf{Q}_p^N \cong \varprojlim_k \mathbf{Q}_p^N / p^k \mathbf{Z}_p^N$; completeness gives surjectivity and $\bigcap_k p^k \mathbf{Z}_p^N = 0$ gives injectivity. For an affinoid perfectoid open $U \subset P_0$, all the spaces $P_k \times_{P_0} U$ are affinoid and finite étale over U . Their inverse limit is an affinoid perfectoid space, pro-étale over U , by [4, Proposition 6.5 and Definition 7.8]. These spaces represent the open restrictions $P \times_{P_0} U$ and hence glue to represent P . The morphism $P \rightarrow P_0 \rightarrow B$ is pro-étale. Finally, the constructions on the original affinoid open cover of B represent restrictions of the same v -sheaf P , so they glue over B as well. $\square$

Corollary 3.4. *If $W_{n-1,d-1}$ is represented by a perfectoid space, then U_t is represented by a perfectoid space.*

Proof. By Proposition 3.1, the base $\mathcal{V}_t$ is perfectoid. By Proposition 3.2, U_t is a torsor over $\mathcal{V}_t$ under the F -local system $\mathcal{H}_t$. Apply Lemma 3.3. $\square$

4 Transverse global sections

Lemma 4.1 (A fixed supply of global transverse sections). *Assume $d > 1$. There is a short exact sequence on X*

$$0 \longrightarrow A \longrightarrow V \longrightarrow \mathcal{O}_X(1/(d-1)) \longrightarrow 0, \quad A \cong_X \mathcal{O}_X^{\oplus(r+1)}. \quad (6)$$

Choose a decomposition

$$A = L_0 \oplus_X \cdots \oplus_X L_r, \quad L_i \cong_X \mathcal{O}_X,$$

and let $t_i : \mathcal{O}_X \xrightarrow{\sim} L_i \hookrightarrow V$ be the resulting stably injective sections. For every point $x \in |W_{n,d}|$, at least one t_i is transverse to the kernel at x , equivalently $x \in |U_{t_i}|$. More precisely, if $\xi : S_x = \mathrm{Spa}(K, K^+) \rightarrow W_{n,d}$ represents x , with K any perfectoid affinoid field, then ξ factors through some U_{t_i} .

Proof. Apply [1, Theorem 1.1.2] with

$$F_1 = \mathcal{O}_X^{\oplus(r+1)}, \quad F_2 = \mathcal{O}_X(1/(d-1)), \quad E = V = \mathcal{O}_X(1/n).$$

The ranks and degrees agree, $0 < 1/(d-1)$, and the straight Harder–Narasimhan polygon of V lies below that of $F_1 \oplus F_2$, so (6) exists. The inclusion $A \hookrightarrow V$ has vector-bundle cokernel and each $L_i \hookrightarrow A$ is split. The argument in Lemma 2.2 therefore proves that all t_i are stably injective.

Fix a representative $\xi : S_x = \mathrm{Spa}(K, K^+) \rightarrow W_{n,d}$ of x ; no algebraic closedness assumption on K is made. Write $K_x \hookrightarrow V_x$ and $Q_x = V_x/K_x$ for its kernel and quotient on X_{S_x} . In particular, K_x has rank r and degree 0. Pulling back (6) preserves exactness. If every composite

$$L_{i,x} \longrightarrow V_x \longrightarrow Q_x$$

were zero, then $A_x \rightarrow Q_x$ would vanish, so the injection $A_x \hookrightarrow V_x$ would factor through K_x . This would inject a rank- $(r+1)$ vector bundle into a rank- r vector bundle, which is impossible. Hence some $L_{i,x} \rightarrow Q_x$ is nonzero.

For a point S_x of arbitrary rank, restriction to $X_{\mathrm{Spa}(K, K^\circ)}$ is an equivalence on coherent sheaves by [3, §A.2]. We may therefore check injectivity on this integral rank-one curve. A nonzero map from a line bundle to a vector bundle is injective there, so the sum map

$$E_x := K_x \oplus L_{i,x} \longrightarrow V_x$$

is injective. To obtain stable injectivity, observe that E_x has rank $r + 1$ and degree 0, whereas V_x is semistable of slope $1/n$. Since $d > 1$,

$$\frac{\deg(E_x) + 1}{\mathrm{rk}(E_x)} = \frac{1}{r + 1} > \frac{1}{n} = \mu(V_x).$$

Thus [3, Lemma A.2.1], which is stated for arbitrary points, shows that E_x is saturated in V_x , so V_x/E_x is a vector bundle. By the local splitting argument in Lemma 2.2, the sum map is therefore stably injective. Consequently the original morphism ξ factors through U_{t_i} , and in particular $x \in |U_{t_i}|$. $\square$

5 Proof of the theorem

Proof of Theorem 1.2. We prove the assertion simultaneously for all $n > d$ by induction on d . For $d = 1$, Lemma 2.2 identifies $W_{n,1}$ with the functor of subbundles

$$K \hookrightarrow \mathcal{O}_{X_S}(1/n), \quad K \cong_{\mathrm{geom}/S} \mathcal{O}^{\oplus(n-1)}.$$

Johansson–Ludwig–Hansen identify this functor with the quotient of the infinite-level Lubin–Tate space by the parabolic of block type $(n - 1, 1)$ [3, Theorem A.1.1], and this diamond quotient is represented by the perfectoid space $\mathcal{M}_{P(K)}$ by [3, Theorem A and Proposition 4.2.7].

Now let $d > 1$ and assume that $W_{n-1,d-1}$ is perfectoid. By Proposition 2.3, $W_{n,d}$ is a diamond. Let $x \in |W_{n,d}|$. Choose the fixed sequence and sections $t_0, \dots, t_r$ of Lemma 4.1. That lemma gives an index i with

$$x \in |U_{t_i}|.$$

The subdiamond $U_{t_i} \subset W_{n,d}$ is open by Lemma 2.4, and it is represented by a perfectoid space by Corollary 3.4 and the induction hypothesis. Thus every point of $W_{n,d}$ has an open perfectoid neighborhood. Lemma 2.1 now implies that $W_{n,d}$ itself is represented by a perfectoid space.

After $d - 1$ induction steps one reaches $W_{n-d+1,1}$, so the proof is complete. $\square$

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